

Prediction of Medical Care Abandonment in Italy Using Population Survey Data

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Abstract

This study investigates the phenomenon of medical care abandonment in Italy using data from the 2019 European Health Interview Survey (EHIS). By applying machine learning techniques, specifically XGBoost and multinomial logistic regression, we aim to identify individuals at risk of forgoing necessary healthcare services. To enhance model robustness, we employed advanced preprocessing, feature selection with LASSO and gain-based importance, and a stability analysis via bootstrap resampling. We further implemented a conformal prediction framework to provide uncertainty-aware predictions with guaranteed statistical coverage. Results show that XGBoost significantly outperforms logistic regression in identifying individuals who avoid care, with superior precision and F1-scores, particularly for minority behavior classes. These findings offer valuable insights for policymakers and healthcare providers seeking to design targeted interventions that reduce inequalities and improve access to care.

1 Introduction

Avoiding necessary medical visits or treatments can lead to poorer health outcomes and increased long-term healthcare costs [Istituto Superiore di Sanità, 2023]. It is therefore important to understand the factors that lead individuals to delay or skip medical care. The 2019 European Health Interview Survey (EHIS), conducted in Italy before the COVID-19 pandemic, provides detailed information on population health, healthcare use, and health determinants such as self-perceived health, chronic conditions, lifestyle habits,

preventive care, and socio-demographic characteristics [Istat, 2021].

Significantly, the EHIS also includes data on difficulties in accessing healthcare services, making it a useful tool for identifying people who were already experiencing barriers to care before the pandemic.

This study uses data from the 2019 EHIS to classify individuals who tend to delay or miss medical care. Using the XGBoost machine learning algorithm, it detects patterns among socio-demographic, health-related, and lifestyle factors associated with unmet healthcare needs. The aim is to offer a clearer understanding of healthcare access issues and support the development of policies that reduce inequalities and ensure fair and effective access to services for all.

2 Sampling Design

The survey was conducted on a sample of approximately 24,000 households residing in 797 Italian municipalities. Each member of the selected households was required to complete an individual questionnaire. Responses were provided directly by the respondents themselves. In cases where the individual was under 15 years of age, a parent or another adult household member answered on their behalf. Additionally, an adult household member could respond in place of the selected individual if the latter was absent for the entire survey period, hospitalized, or affected by a severe permanent health condition [Istat, 2025].

The sampling design followed the general framework commonly adopted in Istat's household surveys. It employed a multi-stage sampling scheme with municipalities and households as the primary stages, stratified by municipalities [Istat, 2022]. Households were randomly selected from the Base Register of Individuals (RBI) according to a sampling strategy aimed at constructing a statistically representative sample of the resident population in Italy [Istat, 2025].

3 Exploratory Data Analysis

The dataset comprises 45,930 observations and 244 variables, encompassing a wide range of demographic, socioeconomic, and health-related information.

The purpose of this exploratory analysis is to provide an initial overview of the data structure, identify key patterns and trends, and highlight relevant relationships that may inform subsequent modeling and interpretation.

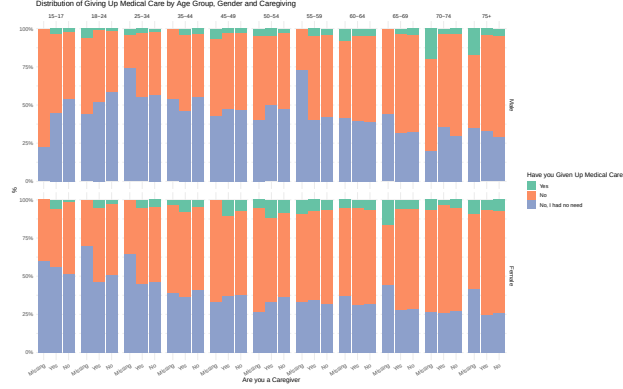


Figure 1: Distribution of Giving Up Medical Care by Age Group, Gender and Caregiving

For instance, Figure 1 clearly shows that the tendency to give up medical care increases with age. Gender emerges as a significant factor: at the same age, women are more likely than men to report unmet healthcare needs, especially during middle age.

This pattern is even more pronounced among individuals who provide care to others. Women between the ages of 40 and 60, who are often responsible for both their children and elderly parents, appear especially vulnerable to neglecting their own health. The figure reinforces the idea that the combined family responsibilities typical of the so-called sandwich generation are a major contributor to the presence of unmet healthcare needs.

It is also important to note that the figure includes missing values for the question regarding caregiving responsibilities. Since many of the respondents with missing data still report having unmet medical needs, it is essential to consider this group in the analysis.

Another critical issue affecting the national health system is the delay in receiving medical care due to either excessively long waiting lists or access problems related to distance and lack of transportation.

In the first chart (Figure 2), which focuses on delays caused by waiting lists, it is clearly evident that individuals who experienced this type of delay show a higher incidence of unmet medical needs compared to those who did not experience delays. This pattern is visible across all geographical areas, but it is particu-

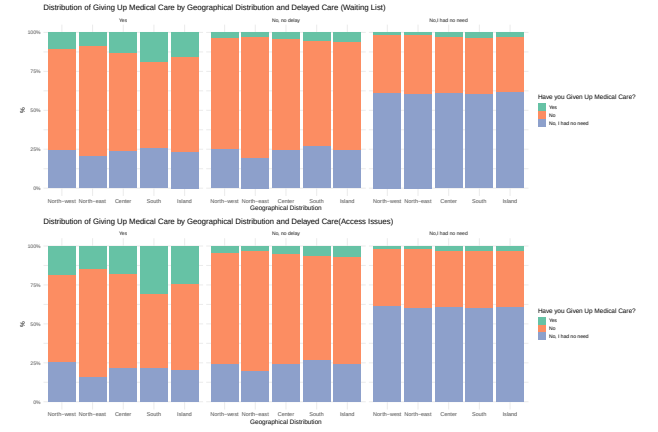


Figure 2: Distribution of Giving Up Medical Care by Geographical Distribution and Delayed Care

larly pronounced in the South and the Islands, where the rate of unmet care is significantly higher than in the North.

The second chart (Figure 2) focuses on delays linked to access difficulties, such as distance or lack of transport. Again, individuals who experienced such delays are more likely to report having given up medical care. However, the geographical gap is even more marked in this case: care avoidance due to access problems is significantly higher in the South and the Islands, suggesting structural inequalities in service availability and accessibility.

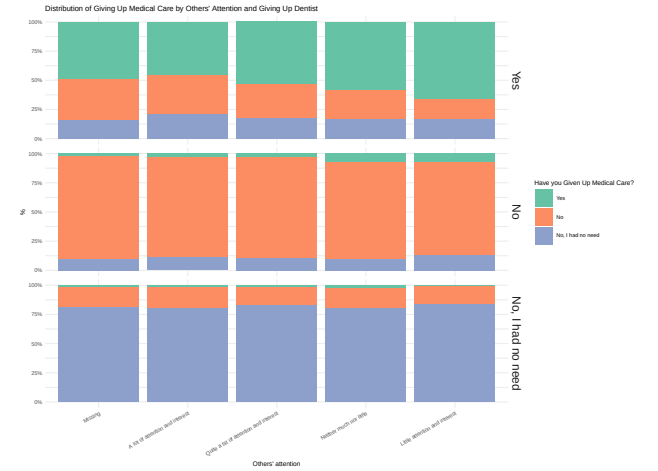


Figure 3: Distribution of Giving Up Medical Care by Others' Attention and Giving Up Dentist

Figure 3 examines the relationship between perceived social attention and the decision to give up medical care, revealing meaningful patterns in healthcare access behaviors.

This figure reveals two main findings: individuals who feel they receive little attention from others are more likely to skip both medical and dental treat-

ments, suggesting that social isolation may hinder access to care due to a lack of support or motivation. In contrast, those with stronger social ties are less likely to avoid care, highlighting the positive role of social engagement in encouraging healthcare utilization.

4 Data Preprocessing and Feature Engineering

The analysis of health survey data required careful preprocessing to handle its inherent complexity. The dataset’s high dimensionality and conditional response patterns necessitated a structured approach to ensure robust results.

We employed regular expressions to systematically group variables by thematic domains (Figure 4), enabling efficient processing of discontinuous question sequences.

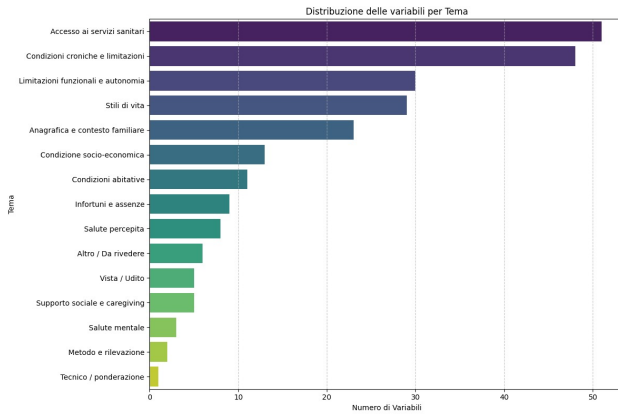


Figure 4: Distribution of Variables by Themes

The survey’s skip logic introduced missing data that required special handling: we distinguished true missing values from logically inapplicable responses, with different treatment strategies for each. However, if they answered negatively, subsequent vaccination-related questions were coded as ”-2” (not available). We excluded proxy responses, cases where someone answered on behalf of the intended respondent, from our analysis. Although the dataset included a proxy indicator variable, we discovered inconsistencies in its application across certain variables. This exclusion resulted in the removal of approximately 2,800 observations but significantly improved the clarity and reliability of our model.

Our preprocessing strategy was deliberately selective. Rather than automatically encoding every categorical variable (which would have generated thousands of dimensions), we implemented a controlled approach that resulted in approximately 400 variables. This number strikes a balance between comprehensive representation and computational efficiency.

This data-driven methodology revealed significant relationships that might have remained hidden using conventional theory-based variable selection approaches. By letting the data guide our feature engineering process, we uncovered novel patterns and insights that enriched our analysis and led to meaningful discoveries.

5 Methodology

The primary objective of our predictive modeling task is to estimate the probability that an individual belongs to a specific class of healthcare behavior, with particular focus on identifying those who renounce medical care. Given the categorical nature of our outcome variable with multiple classes, we employed two complementary classification approaches: Multinomial Logistic Regression and XGBoost (Extreme Gradient Boosting).

5.1 Multinomial Logistic Regression

Multinomial logistic regression (MNL) extends binary logistic regression to problems with multiple outcome categories, making it well-suited for our analysis of healthcare avoidance behavior. Let $Y_i \in \{1, 2, \dots, K\}$ denote the outcome variable for observation i , representing different healthcare utilization patterns, and let $X_i \in \mathbb{R}^p$ be the vector of covariates including socioeconomic, demographic, and health-related factors.

The model defines the conditional probability of an individual i belonging to class k as:

$$P(Y_i = k | X_i) = \frac{\exp(\beta_k^\top X_i)}{\sum_{j=1}^K \exp(\beta_j^\top X_i)}, \quad \text{for } k = 1, \dots, K, \quad (1)$$

where $\beta_k \in \mathbb{R}^p$ represents the parameter vector associated with class k . For identifiability purposes, one class (typically $k = 1$) is designated as the baseline reference category, with its parameter vector constrained to $\beta_1 = \mathbf{0}$.

The model parameters are estimated through maximum likelihood estimation, which involves minimizing the negative log-likelihood function:

$$\mathcal{L}(\beta) = - \sum_{i=1}^n \sum_{k=1}^K \mathbb{I}\{Y_i = k\} \log P(Y_i = k | X_i), \quad (2)$$

where $\mathbb{I}\{Y_i = k\}$ is an indicator function that equals 1 when observation i belongs to class k and 0 otherwise.

5.2 XGBoost Classifier

XGBoost (Extreme Gradient Boosting) [Chen and Guestrin, 2016] is a powerful machine learning algorithm that constructs an ensemble of decision trees sequentially, with each new tree focusing on correcting errors made by previous trees.

The algorithm optimizes a regularized objective function that balances prediction accuracy with model complexity:

$$\mathcal{L}(\Theta) = \sum_{i=1}^n \ell(y_i, \hat{y}_i) + \sum_{t=1}^T \Omega(f_t) \quad (3)$$

The first term measures prediction accuracy using multinomial log loss for multiclass problems:

$$\ell(y_i, \hat{y}_i) = - \sum_{k=1}^K \mathbb{I}\{y_i = k\} \log \hat{p}_i^{(k)} \quad (4)$$

where K represents the number of classes, $\mathbb{I}$ is an indicator function, and $\hat{p}_i^{(k)}$ is the predicted probability that observation i belongs to class k .

The second term controls model complexity through regularization:

$$\Omega(f) = \gamma T + \frac{1}{2} \lambda \sum_{j=1}^T w_j^2 \quad (5)$$

where T is the number of leaves, w_j is the score assigned to leaf j , γ controls the minimum loss reduction needed for splitting, and λ applies L2 regularization to leaf weights.

After constructing M trees, the final prediction combines all tree outputs:

$$\hat{y}_i = \phi(x_i) = \sum_{m=1}^M f_m(x_i) \quad (6)$$

For multiclass classification, predictions are transformed into probabilities using the softmax function:

$$\hat{p}_i^{(k)} = \frac{\exp(\phi_k(x_i))}{\sum_{l=1}^K \exp(\phi_l(x_i))} \quad (7)$$

What distinguishes XGBoost is its efficient optimization technique using second-order approximation of the loss function:

$$\mathcal{L}^{(t)} \approx \sum_{i=1}^n [g_i f_t(x_i) + \frac{1}{2} h_i f_t^2(x_i)] + \Omega(f_t) \quad (8)$$

where g_i and h_i represent the first and second derivatives of the loss function. This approximation allows XGBoost to find optimal tree structures more

efficiently than traditional gradient boosting methods, making it particularly effective for our healthcare avoidance prediction task.

5.3 Model Evaluation Framework

We evaluated our models using both traditional classification metrics and an advanced conformal prediction framework, providing complementary perspectives on model performance and reliability.

5.3.1 Traditional Classification Metrics

Both models output class probabilities for each individual. The predicted class is obtained by taking the argmax over these posterior probabilities:

$$\hat{Y}_i = \arg \max_k P(Y_i = k | X_i) \quad (9)$$

Models were evaluated on a held-out test set using standard classification metrics:

- **Precision:** The proportion of predicted positives that are correct, calculated for each class
- **Recall:** The proportion of actual positives that are correctly identified, calculated for each class
- **F1-Score:** The harmonic mean of precision and recall, providing a balanced measure of model performance

5.3.2 Conformal Prediction Framework

Beyond traditional metrics, we employed conformal prediction [Wieczorek, 2023], a rigorous framework that produces prediction sets with statistical guarantees of coverage. Unlike conventional methods that output single point predictions, conformal prediction generates sets of candidate outcomes that contain the true value with a user-specified confidence level $(1 - \alpha)$.

This approach offers several advantages:

- Provides robust uncertainty quantification
- Makes minimal distributional assumptions
- Maintains validity under distribution shifts
- Offers finite-sample coverage guarantees

5.3.3 Mondrian Inductive Conformal Prediction

We specifically implemented Mondrian inductive conformal prediction [Johansson et al., 2015], which extends the standard framework by conditioning conformity scores on class or feature subgroups. This approach addresses class imbalance and heterogeneity in the data, which is particularly important for our

healthcare avoidance classification where certain behavior classes are underrepresented.

The implementation follows these key steps:

1. Split the data into proper training, calibration, and test sets
2. Train the model (XGBoost or logistic regression) on the proper training set
3. For each class k , compute class-specific nonconformity scores $s_k(x_i, y_i)$ from calibration set probabilities
4. Calculate class-conditional thresholds that ensure the desired coverage level $(1 - \alpha)$ within each class
5. Apply these thresholds to test samples to determine prediction sets

Our specific implementation computes nonconformity scores as $1 - \hat{p}_k$, where $\hat{p}_k$ is the predicted probability for class k . For each class, we determine a threshold using the quantile function:

$$\hat{q}_k = \text{quantile}_{1-\alpha} \{1 - \hat{p}_k(x_i) : y_i = k, x_i \in \mathcal{D}_{\text{cal}}\} \quad (10)$$

A test instance’s prediction set then includes all classes k where the nonconformity score falls below the corresponding threshold:

$$\hat{C}(x_{\text{test}}) = \{k : 1 - \hat{p}_k(x_{\text{test}}) \leq \hat{q}_k\} \quad (11)$$

By partitioning the data space in this manner, Mondrian conformal prediction ensures that coverage guarantees hold within each class independently, providing balanced and locally valid uncertainty quantification even with class imbalance. This approach is particularly valuable for healthcare avoidance prediction, where reliable identification of minority classes (those who renounce care) is crucial for public health interventions.

The performance of conformal prediction is evaluated using two key metrics:

- **Empirical coverage:** The proportion of test samples where the true class is included in the prediction set, which should be at least $1 - \alpha$
- **Average prediction set size:** The mean number of classes included in prediction sets, with smaller sets indicating more informative predictions

This dual evaluation framework—combining traditional classification metrics with conformal prediction—provides a comprehensive assessment of both predictive accuracy and reliable uncertainty quantification, essential for responsible decision-making in healthcare policy.

6 Feature Selection and Stability Analysis

To identify the most influential factors associated with healthcare avoidance from our high-dimensional dataset, we employed two complementary approaches: LASSO regularization for logistic regression and feature importance analysis for XGBoost. Both methods were supplemented with bootstrap resampling to ensure stability and reliability of our findings.

6.1 Feature Selection Methodology

For the multinomial logistic regression, we implemented LASSO (Least Absolute Shrinkage and Selection Operator) [Tibshirani, 1996] regularization, which adds an L1 penalty term to the log-likelihood function:

$$\mathcal{L}_{\text{LASSO}}(\beta) = \mathcal{L}(\beta) + \lambda \sum_{k=2}^K \sum_{j=1}^p |\beta_{kj}| \quad (12)$$

This penalty forces less important feature coefficients to exactly zero, effectively removing them from the model. We calibrated the regularization strength to $C = \frac{1}{\lambda} = 0.001$, balancing model sparsity and predictive performance.

For the XGBoost classifier, we leveraged the algorithm’s native gain-based feature importance scores, which quantify each feature’s contribution to reducing the loss function based on how frequently and effectively variables are used in the ensemble’s decision trees. Unlike linear models, XGBoost can capture complex non-linear relationships and interactions between features without requiring explicit specification.

6.2 Bootstrap Stability Analysis

Initial experiments revealed that single-split validation produced results highly dependent on the particular random seed used for splitting. To mitigate this variability, we implemented a bootstrap resampling [Efron and Tibshirani, 1994] approach with 100 iterations, following this workflow for each sample:

1. Generate a bootstrap sample with replacement from the original dataset
2. Split the sample into training (60%), calibration (20%), and test (20%) sets
3. Apply feature selection on the training data
4. Fit models on the selected features
5. Evaluate performance and record selected features for frequency analysis

The Multinomial Logistic Regression model demonstrated strong explanatory power with a Pseudo R-squared of 0.51 and highly significant likelihood ratio test.

Figure 8 illustrates the odds ratios and standard errors of the coefficients.

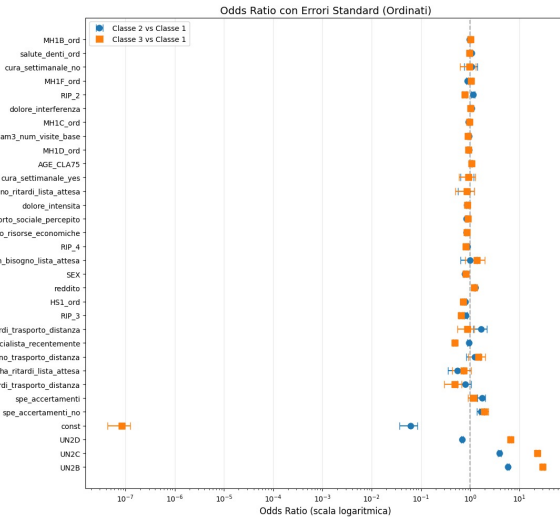


Figure 8: ODDS-Ratio with Standard Error from Logistic Regression Model

The healthcare renunciation indicators showed strong effects with odds ratios ranging from 5 to 30 for Classes 2 and 3. UN2B and UN2C substantially increased the odds of being in these classes, while UN2D showed a more complex pattern, reducing odds for Class 2 but strongly increasing odds for Class 3 ($OR \approx 10$). Access barriers and socioeconomic factors showed moderate but significant effects, highlighting the multifactorial nature of healthcare avoidance.

8.1 Explainable AI and SHAP Analysis of the XGBoost Model

Modern machine learning algorithms such as XGBoost are powerful but often considered “black box” models, since their internal logic and decision boundaries are not easily interpretable. This opacity can be a limitation in sensitive domains like healthcare and policy, where understanding the “why” behind a model’s decision is crucial. **Explainable Artificial Intelligence (XAI)** [Lundberg and Lee, 2017] aims to bridge this gap, providing tools to interpret, visualize, and trust the predictions of complex models.

A key method in the XAI toolkit is **SHAP** (SHapley Additive exPlanations) [Lundberg et al., 2020], which leverages concepts from cooperative game theory to assign each feature a value quantifying its contribution to a model’s prediction. SHAP values provide both global (which features matter overall) and local (why a specific prediction was made) interpretability, making them particularly suitable for tree-based ensemble models like XGBoost.

8.2 Global Feature Importance with SHAP Values

Figure 9 presents the SHAP summary plot for the XGBoost model. Each point represents a SHAP value for an individual observation, colored by the corresponding feature value. Features are ranked by their overall importance (mean absolute SHAP value), offering an intuitive overview of which factors most strongly influence the model’s predictions and in what direction.

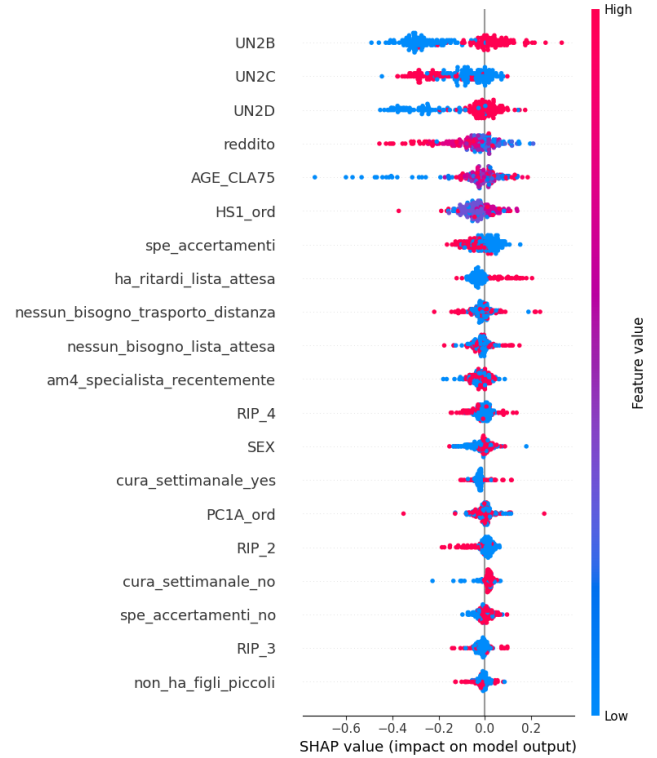


Figure 9: SHAP summary plot showing the global importance and effect of each variable in the XGBoost model.

Interpretation: Figure 9 shows a SHAP summary plot, which provides a global view of how each variable influences the XGBoost model’s predictions.

The SHAP summary plot highlights which factors most strongly drive predictions in the XGBoost model and illustrates how both high and low values for each feature can increase or decrease the likelihood of healthcare avoidance, depending on their SHAP value direction.

8.3 Interpretation of Individual Feature Effects

To further interpret how the most influential variables affect predictions, Figures 10–14 display SHAP dependence plots for the top features. These plots show, for each class, the relationship between the feature value

and its SHAP value—that is, its marginal contribution to the predicted log-odds of each outcome. Each colored dot represents an observation, and the trend lines summarize the average effect for each class. Quartile lines and min/max annotations help to contextualize the distribution and extremity of feature impacts.

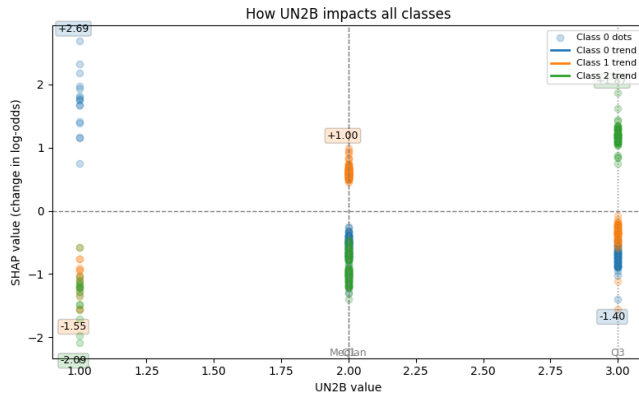


Figure 10: SHAP dependence plot for UN2B: impact on all outcome classes.

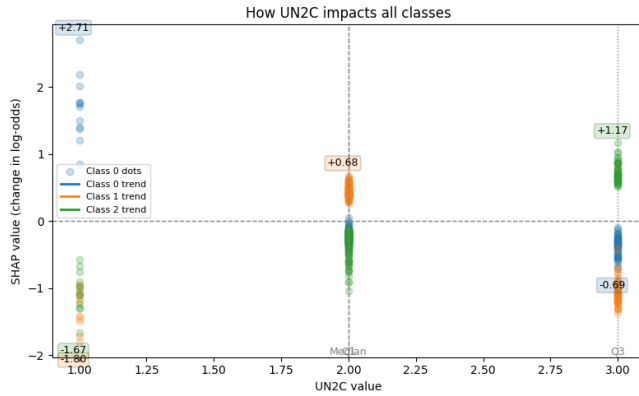


Figure 11: SHAP dependence plot for UN2C: impact on all outcome classes.

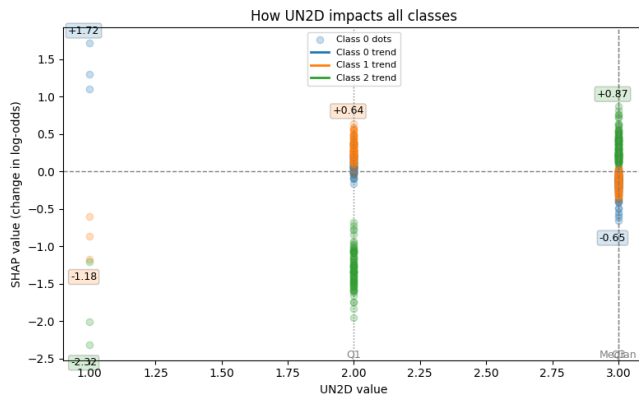


Figure 12: SHAP dependence plot for UN2D: impact on all outcome classes.

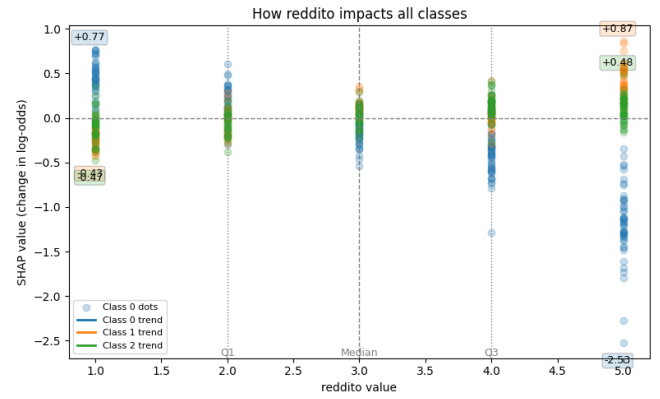


Figure 13: SHAP dependence plot for `reddito`: impact on all outcome classes.

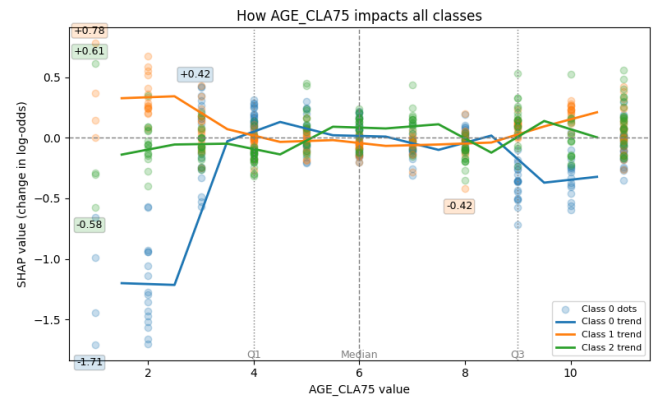


Figure 14: SHAP dependence plot for `AGE_CLA75`: impact on all outcome classes.

Interpretation: The SHAP dependence plots provide nuanced insights into how changes in key variables shift the model’s predictions for different outcome categories. For example, higher values of the “UN2B” and “UN2C” variables are associated with increased model log-odds for certain classes, as indicated by the trend lines. The “reddito” (income) and “AGE_CLA75” (age class) features also demonstrate distinct effects across outcome categories, underlining the importance of individual socioeconomic and demographic characteristics in healthcare avoidance. This allows for a more granular understanding of risk factors and their effects in a multi-class prediction context.

9 Model Performance Comparison

We evaluated model performance using both unweighted and population-weighted metrics. This two-step approach shows how well models perform on the raw data and how results generalize to the broader population.

9.0.1 Unweighted Model Performance

XGBoost outperforms Logistic Regression for identifying healthcare avoiders (Class 1), with notably fewer false positives.

Table 1: Unweighted Confusion Matrix for Logistic

True Class	Predicted Class			Total
	Class 1	Class 2	Class 3	
Class 1	333	72	38	443
Class 2	853	3,429	518	4,800
Class 3	131	329	2,984	3,444
Total	1,317	3,830	3,540	8,687

Table 2: Unweighted Confusion Matrix for XGBoost

True Class	Predicted Class			Total
	Class 1	Class 2	Class 3	
Class 1	341	68	31	440
Class 2	227	4,148	424	4,799
Class 3	77	338	3,028	3,443
Total	645	4,554	3,483	8,682

XGBoost also achieves higher precision and F1-score for healthcare avoiders, with better recall:

Table 3: Classification Report on Test Set for Logistic

Class	Precision	Recall	F1-score
1	0.25	0.73	0.37
2	0.89	0.72	0.80
3	0.85	0.86	0.85

Table 4: Classification Report on Test Set for XGB

Class	Precision	Recall	F1-score
1	0.47	0.76	0.58
2	0.90	0.85	0.88
3	0.87	0.87	0.87

Summary: XGBoost reduces false positives by two-thirds compared to Logistic Regression and increases the F1-score for healthcare avoiders by 57%.

9.0.2 Population-Weighted Model Performance

When adjusting for population representativeness, the advantages of XGBoost are even more pronounced:

- **Correctly identified healthcare avoiders (Class 1):** XGBoost $\approx 387,000$, Logistic Regression $\approx 385,000$
- **False positives for Class 1:** XGBoost 363,000, Logistic Regression 1,087,000
- **F1-score for Class 1:** XGBoost 0.63, Logistic Regression 0.39 (62% improvement)

Table 5: Population-Weighted CM for Logistic

True Class	Predicted Class			Total
	Class 1	Class 2	Class 3	
Class 1	384,680	81,915	47,033	513,628
Class 2	927,268	3,798,555	582,830	5,308,653
Class 3	159,681	368,377	3,436,015	3,964,073
Total	1,471,629	4,248,847	4,065,878	9,786,354

Table 6: Population-Weighted CM for XGBoost

True Class	Predicted Class			Total
	Class 1	Class 2	Class 3	
Class 1	386,717	68,997	26,990	482,704
Class 2	282,244	4,532,658	481,242	5,296,144
Class 3	80,704	368,737	3,515,305	3,964,746
Total	749,665	4,970,392	4,023,537	9,743,594

Table 7: Population-Weighted Report for Logistic

Class	Precision	Recall	F1-score
1	0.26	0.75	0.39
2	0.89	0.72	0.80
3	0.85	0.87	0.86

Table 8: Population-Weighted Report for XGBoost

Class	Precision	Recall	F1-score
1	0.52	0.80	0.63
2	0.91	0.86	0.88
3	0.87	0.89	0.88

9.1 Conformal Prediction Set Evaluation

To assess model uncertainty, we used conformal prediction sets: [Vovk et al., 2005]

- Both models include the true class in their prediction sets about 95% of the time (mean coverage).
- XGBoost’s prediction sets are smaller (mean size 1.29 vs. 1.86), showing higher confidence.

Model	Mean coverage	Mean set size
XGBoost	0.949 ± 0.004	1.29 ± 0.02
Logistic Regression	0.950 ± 0.003	1.86 ± 0.05

Table 9: Bootstrap mean and standard deviation of conformal prediction set coverage and mean set size for each model.

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